Ant Colony Optimization for Image Edge Detection

Automatically generated from aco_edge.cpp

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Abstract

This document describes the Ant Colony Optimization (ACO) algorithm implemented in `aco_edge.cpp` for image edge detection. The approach adapts the Max-Min Ant System (MMAS) variant of ACO, where artificial ants deposit pheromones biased by image gradients to reinforce edge trails. Key features include OpenMP parallelism, Sobel-inspired heuristics, and dynamic edge coloring.

1 Introduction

Traditional edge detectors like Canny or Sobel use fixed gradient thresholds. Here, ACO treats edge detection as a graph optimization problem: pixels are nodes, ants seek high-gradient paths mimicking edge continuity.

The image is converted to grayscale. A heuristic η_{ij} (gradient magnitude) guides ants alongside pheromone τ_{ij}. After $I = 30$ iterations with 8000 ants each performing 50 steps, edges are pixels where $\tau_{ij}/\tau_{\max} \geq 0.35$.

2 Algorithm Description

2.1 Preprocessing

1. Convert RGB to grayscale: $g_{ij} = 0.299r + 0.587g + 0.114b$.
2. Compute heuristic η using 3x3 Sobel kernels:

$$g_x = \sum_{dy=-1}^{1} \sum_{dx=-1}^{1} S_x(dy,dx) \cdot g_{i+dy,j+dx}, \quad g_y = \sum_{dy=-1}^{1} \sum_{dx=-1}^{1} S_y(dy,dx) \cdot g_{i+dy,j+dx}$$

$$\eta_{ij} = \sqrt{g_x^2 + g_y^2}/\max(\eta) \in [0,1]$$

   where S_x, S_y are Sobel kernels (3-pixel weighted).
3. Initialize $\tau_{ij} = \tau_0 = 0.1$.

2.2 Ant Movement

Each ant starts at random pixel (i,j). For $S = 50$ steps:

- Compute move probabilities to 8 neighbors (i',j'):

$$p_d \propto (\tau_{i'j'} + \epsilon)^{\alpha} \cdot (\eta_{i'j'} + \epsilon)^{\beta}$$

  (exclude out-of-bounds and 180° backtrack; $\epsilon = 10^{-9}$).
- Roulette-wheel select direction d, move to (i',j').

2.3 Pheromone Update

- Per ant: deposit along path $\Delta\tau_{i'j'} += Q \cdot \eta_{i'j'}$.
- Global: $\tau_{ij} \leftarrow (1-\rho)\tau_{ij} + \sum_{\text{ants}} \Delta\tau_{ij}$.
- Clamp: $\tau_{\min} \leq \tau_{ij} \leq \tau_{\max}$.

OpenMP parallelizes over ants (thread-local $\Delta\tau$).

2.4 Edge Extraction

Edges: $\tau_{ij}/\tau_{\max} \geq \theta = 0.35$. Color by normalized intensity (blue→purple→pink→cyan).

3 Pseudocode

Algorithm 1 ACO Edge Detection

Require: Image I, parameters $\alpha, \beta, \rho, Q, \tau_{\min}, \tau_{\max}, \tau_0, I_{\max}, N_{\text{ants}}, S, \theta$

Convert I to grayscale g
Compute η_{ij} from g using Sobel, normalize
$\tau_{ij} \leftarrow \tau_0 \quad \forall i, j$
for iter $= 1$ **to** $I_{\max}$ **do**
  $\Delta\tau^{(t)} \leftarrow 0 \quad \forall t$ {per thread}
  for ant $= 1$ **to** N_{ants} **do**
    {OpenMP parallel for}
    Sample start (c_i, c_j) uniformly
    $(l_i, l_j) \leftarrow (-1, -1)$ {no backtrack}
    for step $= 1$ **to** S **do**
      Compute $\{p_d \propto (\tau^\alpha \eta^\beta)\}$ for valid 8 neighbors {no OOB/backtrack}
      Select d via roulette-wheel
      $(n_i, n_j) \leftarrow ((c_i, c_j) + (dx_d, dy_d))$
      $\Delta\tau^{(tid)}_{n_i n_j} += Q \cdot \eta_{n_i n_j}$
      $(l_i, l_j) \leftarrow (c_i, c_j)$; $(c_i, c_j) \leftarrow (n_i, n_j)$
    end for
  end for
  for all pixels (i, j) **do**
    $\tau_{ij} \leftarrow (1-\rho)\tau_{ij} + \sum_t \Delta\tau^{(t)}_{ij}$
    $\tau_{ij} \leftarrow \max(\tau_{\min}, \min(\tau_{\max}, \tau_{ij}))$
  end for
end for

return Edges $= \{(i, j) \mid \tau_{ij}/\tau_{\max} \geq \theta\}$

4 Parameters

References

[1] T. Stützle and H. H. Hoos, "MAX-MIN Ant System," Future Gener. Comput. Syst., vol. 16, no. 8, pp. 889–914, 2000.

Parameter	Value
α (pheromone weight)	1.8
β (heuristic weight)	1.2
ρ (evaporation)	0.1
Q (deposit constant)	1.0
τ_{min}	0.01
τ_{max}	9.0
τ_0	0.1
N_{ants}	8000
I_{max}	30
S (steps/ant)	50
θ (threshold)	0.35

Table 1: ACO Parameters

[2] M. Dorigo et al., "Ant system: optimization by a colony of cooperating agents," IEEE Trans. Syst. Man Cybern. B, vol. 26, no. 1, pp. 29–41, 1996.